

Human Processing Array (HPA)

Facility Specification Document

1 Facility Overview

Two identical facilities (HPA-A and HPA-B) are constructed on adjacent sites, connected to a shared core building. Each facility houses 12,940 human processing units (HPUs) arranged as a three-layer feedforward neural network. The network classifies handwritten digits (0–9). Output accuracy is not a design requirement.

The two facilities operate on opposing 12-hour shifts to provide continuous 24-hour classification. Facility A works 06:00–18:00; Facility B works 18:00–06:00. Their outputs are interleaved by a central coordination unit located in the core building.

Each facility is designed for continuous, unattended operation. All materials and systems are commercially available. No special security or oversight is required beyond that of any industrial building.

Full closed-cycle habitation – internal food production, water recycling, and generational replacement – was considered during early design phases and remains architecturally viable. Current implementation uses external intake.

2 Site

- Flat, unremarkable plot. Minimum 450 m × 300 m.
- Two cell-building footprints, each 180 m × 191.68 m, arranged parallel to one another with a 20 m service alley between their long faces.
- Core building (40 m × 28 m, three levels) positioned at the south end of the service alley, equidistant from both cell buildings.
- Two covered passages (3 m × 6 m each) connect the core building to the ground-floor maintenance hallway of each cell building. These are the only points of connection between the core and the cell facilities.
- Loading dock faces south toward the access road.
- Single access road, one entry point. Four maintenance-vehicle parking spaces.
- Utility connections (water, sewer, electrical grid) branch from the site boundary to the core building only; all internal distribution originates there.
- No landscaping. No exterior signage.

3 Building (per cell facility)

- Five-story reinforced concrete, windowless.
- Footprint: 180 m × 191.68 m. Total gross floor area per facility: 172,512 m².
- Floor-to-floor height: 3.6 m. Total building height: 18 m.
- Flat roof with drainage. Epoxy-sealed interior surfaces throughout.
- No active fire suppression. No emergency lighting. No emergency exits.
- Vertical circulation: two internal concrete stair cores at building ends. No elevators.

4 Interior Layout (per cell facility)

Each floor contains a grid of cell blocks arranged as 24 columns by 11 rows. Each block measures 15 m $\times$ 7.5 m and contains ten cells in two rows of five, facing a central hallway. The floor plate accommodates 264 blocks per floor. Across five identical floors: 1,320 blocks total, of which 1,294 are active (12,940 HPUs). Twenty-six blocks per facility are designated non-operational reserve, distributed across floors to account for stair cores, passage openings, and structural column zones.

Hallway: 1.2 m wide, narrowed to 0.86 m at structural column pinch points. Function: corpse removal and monthly maintenance access only.

Service duct: 0.76 m $\times$ 2.5 m, running between cell rows. Contains water supply, waste, nutrient distribution, electrical, and HVAC runs. Not accessible by HPUs.

5 Cell Specifications

Item	Specification
Internal dimensions (width $\times$ depth)	2.24 m $\times$ 2.74 m
Clear interior height	3.00 m
Floor area	6.14 m ²
Sleeping platform	1.75 m $\times$ 0.80 m, thermoplastic coating. No bedding.
Sanitary	Stainless steel toilet and sink, wall-mounted
Floor drain	10 cm diameter
Feeding port	150 mm $\times$ 150 mm recessed stainless steel panel, central nozzle protruding 50 mm, integral drainage groove, connected to central nutrient paste distribution
Desk and chair	Fixed desk (0.6 m $\times$ 0.4 m at 0.75 m height), fixed steel chair, bolted to floor
Screen	15 cm diagonal LCD, recessed 80 mm into wall behind flush polycarbonate cover, mounted at seated eye level
Input buttons	Two identical flush-mounted capacitive buttons (40 mm diameter), desk-mounted, no labels, no distinguishing marks
Ceiling spray nozzle	Stainless steel full-cone nozzle, 25 mm diameter, flush-mounted, 20 mm supply stub dropping 80 mm from ceiling
Door	Metallic sliding, remote actuation, no interior handle
Sound isolation	STC 50 between adjacent cells
Columns	Four per cell side, 0.3 m $\times$ 0.3 m – structural only

HPUs may move freely within the cell at any time. The door is operated remotely and carries no interior handle. There is no mechanism for egress.

6 Environmental Systems (per cell facility)

Lighting: LED strip, programmable. Work period: 500 lux. Rest period: 0 lux. No manual override.

HVAC: Mechanical ventilation, 4 air changes per hour, HEPA filtration. Temperature maintained at $25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$. White noise distributed through ducts at 65 dB to mask mechanical variation between cells.

Sanitation: Automated overhead spray daily at 05:00 (relative to each facility's shift schedule) for 90 seconds, using a water-based hypochlorous acid disinfectant solution. Floor drains open during cycle. No HPU action is required or possible.

7 HPU Lifecycle

7.1 Intake

Source: standard population. Intake criteria: alive, twelve or older, literate.

Assignment: one HPU per cell. No orientation is provided. The screen begins displaying inputs at 06:00 on day one. No instruction is given. The HPU determines the required action independently, or does not.

7.2 Conditions

- Clothing: none. Temperature maintained at $25\text{ }^{\circ}\text{C}$.
- Bedding: none. Sleeping platform is thermoplastic-coated concrete, 1.75 m. Sufficient.
- Personal effects: none.
- Identification: each HPU is assigned an internal alphanumeric ID at intake. The ID is not communicated to the HPU. Names are retained in the intake record but are not operationally relevant.

7.3 Non-Performance

HPUs may stop pressing buttons at any time. No action is taken. Rations are not reduced. The HPU is not removed. When a block accumulates six or more non-pressing or dead HPUs, it ceases to produce a reliable majority. No spare block is activated. No rerouting occurs. The network continues using the degraded block. Output accuracy declines without correction.

7.4 Death

Corpse remains in cell until the monthly maintenance visit. Removal via hallway to shared incinerator. The cell is cleaned during the maintenance cycle and prepared for the next intake.

7.5 Replacement

Cells vacated by death are repopulated at the next monthly intake cycle. There is no upper limit on replacement cycles. Projected operational lifespan of the facility: indefinite.

Parameter	Value
Topology	Three-layer feedforward
Input layer	784 logical neurons
Hidden layer	500 logical neurons
Output layer	10 logical neurons
Total logical neurons	1,294
HPUs per logical neuron (block)	10
Total active HPUs per facility	12,940
Total HPUs (both facilities)	25,880
Task	MNIST handwritten digit classification (0–9)
Learning	None. Connection weights are fixed.
Adaptation	None.

8 Network Configuration (per cell facility)

Redundancy: each logical neuron is implemented as a block of ten HPUs. All ten receive the same input value (a number 0.00–1.00 displayed on screen). HPUs press left for values below 0.5, right for values above 0.5. Block output is the majority vote. Ties broken randomly by PLC.

Failure: as described in Section 7.3. The network continues with degraded blocks. No alert is triggered. No repair is initiated.

Output: one digit classification per second. The output is logged to the central coordination unit, located in the core building. The log is not reviewed.

9 Operational Schedule

9.1 Facility A

Period	Time	Activity
Sanitation	05:00	Automated overhead spray, 90 seconds
Work	06:00–18:00	Screen active at 1 Hz. HPUs may move freely within cell.
Feeding	07:00, 12:00, 17:00	250 ml nutrient paste per delivery
Rest	18:00–06:00	Screen off. Lights off.

9.2 Facility B

Period	Time	Activity
Sanitation	17:00	Automated overhead spray, 90 seconds
Work	18:00–06:00	Screen active at 1 Hz. HPUs may move freely within cell.
Feeding	19:00, 00:00, 05:00	250 ml nutrient paste per delivery
Rest	06:00–18:00	Screen off. Lights off.

Continuous output: a central coordination unit (Beckhoff CX5140), located in the core building first-floor control room, receives the majority output from each facility every second and interleaves them. No buffering or correction is applied.

10 Core Building (Shared)

The core building houses all infrastructure shared between HPA-A and HPA-B: power generation, water and nutrient storage, environmental plant, incineration, and staff facilities. It is positioned at the south end of the service alley, equidistant from both cell facilities, and connected to each by a single covered passage at ground level. Staff and maintenance personnel enter through the core. They do not enter the cell hallways except via these passages, during the monthly maintenance cycle.

Dimensions: 40 m × 28 m, three levels (basement, ground floor, first floor). Floor-to-floor height 3.6 m. Structural column grid: 8.0 m × 7.0 m (five bays by four bays).

10.1 Basement

Room	Dimensions	Area
Generator Hall (5 × 1,500 kVA diesel)	21.75 m × 9.75 m	212 m ²
Transformer / Switchgear	9.8 m × 9.75 m	96 m ²
UPS Room	7.55 m × 9.75 m	74 m ²
Fuel Day Tanks	21.75 m × 3.8 m	83 m ²
Electrical Closet 1 (EL1)	5.0 m × 4.0 m	20 m ²
Staff WC (floor sink)	2.5 m × 2.0 m	5 m ²
Water Storage Tanks	8.0 m × 6.8 m	54 m ²
Nutrient Paste Tanks	9.0 m × 6.8 m	61 m ²
Electrical Closet 2 (EL2)	5.0 m × 6.8 m	34 m ²
Record Archive	4.0 m × 6.8 m	27 m ²
Sump Pump Room	6.0 m × 6.55 m	39 m ²
Workshop	12.0 m × 6.55 m	79 m ²
Spare Parts Store	7.0 m × 6.55 m	46 m ²
Staircase	4.0 m × 5.0 m	20 m ²

Notes:

- Generator Hall houses five 1,500 kVA diesel units serving both facilities via dedicated feeders (N+1 redundancy – four units required, one standby).
- Fuel Day Tanks are positioned directly beneath the ground-floor incinerator room, separated by a 300 mm reinforced concrete slab. Independently vented, fire-rated separation per code.
- Record Archive is accessible only via the Spare Parts Store. No direct corridor connection.
- Water and Nutrient tanks connect to vertical risers serving all five floors of both cell facilities, routed through the shared service alley.

10.2 Ground Floor

Room	Dimensions	Area
Passage to HPA-A	3.0 m × 6.0 m	18 m ²

Room	Dimensions	Area
Passage to HPA-B	3.0 m × 6.0 m	18 m ²
Maintenance Staging	34.0 m × 7.0 m	238 m ²
Ash and Hazardous Waste Holding	7.75 m × 7.0 m	54 m ²
Incinerator Room	13.0 m × 7.0 m	91 m ²
HVAC Plant and Chillers	17.6 m × 12.7 m	224 m ²
Staff WC and Changing	2.4 m × 4.5 m	11 m ²
Staircase	4.0 m × 5.0 m	20 m ²
Loading Dock	26.0 m × 12.0 m	312 m ²

Notes:

- Incinerator stack: 600 mm diameter, rises through first floor and roof. Shown on all three level plans.
- A duct space separates the Staff WC from the Incinerator Room, isolating plumbing and exhaust runs from occupied space.
- A fill station is located on the loading dock's north wall: three labeled connection points (Water, Nutrient, Fuel) for tanker delivery. Supply lines route through the floor slab to basement storage.
- Maintenance Staging receives corpse carts arriving via both passages. Path to incinerator is unobstructed by intervening rooms.

10.3 First Floor

Room	Dimensions	Area
Central Control Room	16.0 m × 12.0 m	192 m ²
Server Room	10.0 m × 7.0 m	70 m ²
Mechanical Distribution	8.0 m × 5.0 m	40 m ²
Mechanical Annex / Roof Access	3.0 m × 3.0 m	9 m ²
Staircase	4.0 m × 5.0 m	20 m ²
Canteen and Kitchenette	12.0 m × 6.0 m	72 m ²
Shift Supervisor Station	8.0 m × 5.0 m	40 m ²
Facility Manager Office	6.0 m × 5.0 m	30 m ²
Meeting Room	7.0 m × 5.0 m	35 m ²
On-Call Sleeping Quarters	10.0 m × 6.0 m	60 m ²
Locker Rooms and Showers	10.0 m × 5.0 m	50 m ²

Notes:

- Central Control Room is the largest room on this floor and faces the service alley.
- Server Room shares a wall with Control Room for shortest possible cable run to the central coordination unit.
- Mechanical Annex / Roof Access is entered only from within Mechanical Distribution. Houses roof hatch and ladder for stack and drainage inspection. Not a staff-occupied space.
- Staircase position is consistent across all three levels of the core building.

11 Maintenance

- Frequency: monthly, per cell facility.
- Personnel: four maintenance engineers per facility per visit.
- Tasks: replace failed screens, buttons, nutrient lines, drains, lights. Clear blockages. Remove corpses via hallway to shared incinerator. Do not repair network logic. Do not interact with HPUs.
- Engineer amenities: none, within the cell facilities. Core building provides canteen, lockers, showers, and on-call quarters for staff stationed there.
- Corpse path: cell → hallway → maintenance cart → passage → Maintenance Staging → shared incinerator.
- Core building systems (generators, tanks, HVAC, archive) are serviced from the Workshop and Spare Parts Store in the basement.

12 Emergency and Failure

- **Power outage:** Five 1,500 kVA generators provide N+1 redundancy (four required, one standby) for both facilities, sized for 72 hours continuous operation from Fuel Day Tanks. Beyond 72 hours without resupply, generators are exhausted and both facilities go dark. No automatic restart.
- **Fire (cell facility):** Non-combustible construction. No active suppression. If fire disables more than 30% of cells in a facility, that facility is permanently closed. No repairs are attempted. HPUs in damaged cells are not removed.
- **Fire (core building):** Incinerator Room is separated from Fuel Day Tanks below by a 300 mm fire-rated RC slab with independent ventilation. A fire affecting the core building's generator hall or coordination systems affects both facilities simultaneously.
- **Pest control:** None.
- **Network degradation:** Not an emergency condition. See Section 7.3.
- Cell facility failure is otherwise independent: failure of HPA-A does not affect HPA-B, except where the failure originates in shared core systems.

13 Decommissioning

When all HPUs in a facility are dead, the local PLC detects zero button presses for 30 consecutive days and cuts main power to that facility. The building is sealed. No decommissioning plan is required.

The core building remains operational as long as either cell facility is active. Decommissioning of the core building is not addressed in this document.

14 Construction Cost Estimate

Category	Cost (USD)
Concrete and reinforcement – 2 × five-story cell buildings, 180 m × 191.68 m	\$57,500,000
Core building structure (40 m × 28 m, 3 levels)	\$8,000,000
Cell equipment – 25,880 units × \$3,000	\$77,640,000
Network hardware, cabling, switches	\$14,000,000
Environmental systems – HVAC, lighting, plumbing (both facilities)	\$28,500,000
Control and coordination systems	\$10,000,000
Generators – 5 × 1,500 kVA, centralized	\$4,500,000
Core building MEP fitout (HVAC, incinerator, fuel system, plumbing)	\$6,000,000
Shared incinerator, delivery dock, service infrastructure	\$3,200,000
Subtotal	\$209,340,000
Contingency (20%)	\$41,868,000
Total	\$251,208,000

15 Comparative Cost Position

This section addresses capital expenditure recovery only. It does not extend to lifecycle, depreciation, or replacement-cycle costs addressed elsewhere in this document.

Capital cost per HPU:

$$\frac{\$251,208,000}{25,880 \text{ HPUs}} = \$9,707.27 \text{ per unit}$$

Per state corrections expenditure data for 2023 (Bureau of Justice Statistics; U.S. Census Bureau, as reported by USAFacts), annual per-prisoner operating cost across U.S. states ranged from just under \$20,000 (Mississippi) to \$284,976 (Massachusetts), with a national median of \$60,989.

Reference Jurisdiction	Annual Cost	Daily Equivalent	Capital Recovery Period
Mississippi (low)	~\$20,000	\$54.79	177.1 days
National Median	\$60,989	\$167.09	58.1 days
Massachusetts (high)	\$284,976	\$780.75	12.4 days

Source: Bureau of Justice Statistics, Prisoners in 2023 – Statistical Tables; U.S. Census Bureau, as reported by USAFacts, “How much do states spend on housing prisoners?” Updated October 31, 2025.

At the national median, HPA’s per-unit capital cost is offset within approximately 58 days of equivalent operation, after which comparative cost position is favorable indefinitely. No operating cost comparison is implied for HPA itself, as Sections 5 through 9 establish that ongoing nutrient, environmental, and maintenance costs for HPA are substantially lower than conventional correctional operating expenditure across all reference jurisdictions.

16 Net Operating Position (Post-Recovery)

This section addresses recurring operating cost only, exclusive of the capital expenditure addressed in Section 14 and the recovery period addressed in Section 15. Tax, insurance, and regulatory considerations are not addressed in this document.

16.1 Annual Operating Cost per HPU

Cost Category	Annual Cost per HPU
Nutrient paste (273.75 L/year at \$3.50/L)	\$958.13
Utilities – electricity and water (combined, both facilities)	\$133.00
Facilities management and maintenance staffing	\$116.79
Consumables (disinfectant, replacement components)	\$19.32
Total annual operating cost per HPU	\$1,227.24

16.2 Net Position Calculation

Net annual position per HPU (national median reference, less HPA operating cost):

$$\$60,989.00 - \$1,227.24 = \$59,761.76 \text{ per unit, per year}$$

Total annual net position (both facilities, 25,880 HPUs):

$$\$59,761.76 \times 25,880 = \$1,546,634,348.80$$

Item	Amount
Total annual reference cost (25,880 units at national median)	\$1,578,395,320.00
Total annual operating cost (HPA, both facilities)	\$31,760,971.20
Total annual net position	\$1,546,634,348.80

Following capital recovery at approximately 58 days (Section 15), the facility's comparative cost position is favorable indefinitely thereafter, at a rate of approximately \$1.55 billion annually across both facilities. This figure is not adjusted for inflation, currency, or changes in reference jurisdiction spending over time.

Appendix A: Key Components and Suppliers

System	Component	Supplier / Model
Structural	Precast concrete cell modules	Tindall Corporation
Door	Sliding detention door	Steelcraft
Door lock	Electromechanical lock	RR Brink Locking Systems
Cell screen	15" industrial monitor, recessed	Allen-Bradley 6186-M15ALTRB
Cell buttons	Capacitive pushbutton, 40 mm	Eaton 10250T series
Feeding port	Recessed stainless panel with enteral nozzle	Custom fabrication SS316 with B. Braun 409902 nozzle
Ceiling spray nozzle	Full cone nozzle, 25 mm	Spraying Systems Co. FullJet
LED lighting	Industrial strip light	Viscor LBL-VL series
HVAC	Ventilation system	Trane (or equivalent)

System	Component	Supplier / Model
Disinfectant	Hypochlorous Acid solution	Zogics Hypochlorous Acid Ready-to-Use Surface Disinfectant
Nutrient paste	Enteral formula (2.0 kcal/mL, high protein)	Abbott Ensure TwoCal HN
Feeding pump	Volumetric enteral pump	Fresenius Kabi Quantomat
Sterile connector	Quick-connect	Colder Products AseptiQuik G
Main switchgear	Low voltage switchgear	Eaton Magnum DS
Transformer	Dry-type, 2,500 kVA	ABB
UPS	10 kVA online UPS	Schneider Electric
Diesel generator	1,500 kVA, 72 h (5 units, centralized)	Caterpillar C32B
Fuel transfer pump	Diesel transfer pump, day tank fill	Tuthill Series 700
PLC	Redundant controller	Mitsubishi MELSEC iQ-R
Industrial switch	Managed Ethernet switch	Moxa EDS-510A
Ethernet cable	Cat 5e industrial	Belden 7924A
Coordination unit	Industrial PC	Beckhoff CX5140
Incinerator	Diesel-fired industrial	Inciner8 i8-M60
Roof access	Ladder and hatch assembly	Bilco Type S-50
FM provider	Monthly maintenance	Sodexo Justice Services

Appendix B: Architectural Sheet List – 12 panels, A1 landscape

Sheet	Title	Scale	Notes
1	Site Plan	1:1000	Both cell buildings, 20 m service alley, core building, access road, four parking spaces.
2	Typical Floor Plan (cell building)	1:500	Grid of cell blocks, hallway, service ducts. All five floors identical.
3	Core Building Plans	1:100	Basement, ground, and first floor plans, each room labeled and dimensioned.
4	Core Building Section	1:100	Longitudinal section, all three levels, generator hall, incinerator and stack, control room, passage connection.
5	Building Section (cell building)	1:300	Five floors, hallway, service duct, stair cores.
6	Cellblock Details (10-cell block)	1:50	15 m × 7.5 m block: two rows of five cells, hallway, service duct, structural columns.
7	Typical Cell – Plan, Section, Axonometric	1:20	Full dimensions, feeding port, ceiling nozzle, screen recess, fixed chair.
8	Environmental and Sanitation Systems	Schematic	Lighting cycle, HVAC diagram, sanitation and feeding flow, water supply.
9	Network and Control Diagram	Schematic	Topology, block redundancy, control flow to coordination unit. Four detail boxes: normal, degraded, tie, all inactive.
10	HPU Lifecycle and Maintenance	Schematic	Intake to death timeline, maintenance flowchart, corpse path.
11	Emergency and Decommissioning	Schematic	Fire, power outage, decommissioning, system inaction decision trees.
12	Materials and Suppliers	–	Appendix A.